

08 - Test and Calibration - Rev B2

Bring-up and production tests

Step	Procedure	Pass criterion
1	Inspect Q1 orientation, RG value, FB1 fitment, and absence of PPTC/TVS	Assembly matches Rev B2 schematic.
2	With power off, verify no continuity between GND_HOST and GND_ISO.	Open circuit / high resistance.
3	Apply +5 V with current-limited supply. Measure voltage before and after Q1 MOSFET.	Q1 MOSFET remains small; PS1 input remains within valid range.
4	Apply reverse polarity briefly using a current-limited supply.	No power reaches PS1; no damage or overheating.
5	Measure 5V_ISO with sensor disconnected and connected.	Rail remains stable; fit RPL1 only if minimum-load behavior required.
6	Set VIN to 0 V and 5 V in each SW1 mode.	Outputs are near 0/2.4 V, 0/3.0 V, and 0/4.5 V respectively.
7	Measure output noise and PS1 ripple with FB1 fitted.	Noise satisfies the product target; compare against 0 Ω only during production.
8	Store zero and full-scale calibration values for each unit or selected mode.	Firmware reconstructs the 0-5 V sensor input correctly.

Calibration formula

For each mode, record VZERO at VIN = 0.000 V and VFULL at VIN = 5.000 V. The reconstructed input voltage is $VIN = (VMEAS - VZERO) \times 5.0 / (VFULL - VZERO)$. This calibration belongs to the analog isolator and is separate from the sensor's physical calibration.

Safety boundary

Rev B2 does not claim surge immunity from an on-board TVS and does not provide a self-resetting input fuse. Use it with a controlled 5 V source and short wiring, or add protection at the system level.